Day - 20
File Handling
in Python

SKILL COURSE

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What is File Handling?

- **Layman Explanation**
- File handling means:
  - Opening a file
  - Reading data from a file
  - Writing data into a file
  - Closing a file
- Just like you open a notebook, write something, read something, and then close it.

- **Technical Explanation**
- File handling in Python uses the built-in function:

```
open(filename, mode)
```

Modes:

Mode	Meaning
"r"	read
"w"	write (overwrite)
"a"	append (add new content)
"r+"	read + write
"w+"	write + read

Why File Handling is Important for Data Analysts?

1. Read log files
2. Read data from text reports
3. Clean unwanted characters
4. Store cleaned results
5. Save output for automated dashboards
6. Create summary reports
7. Automate repetitive file tasks

Opening & Reading a Text File

```
file = open("sample.txt", "r")
content = file.read()
print(content)
file.close()
```

Read Line by Line

```
file = open("sample.txt", "r")

for line in file:
    print(line.strip())

file.close()
```

Read First N Characters

```
file = open("sample.txt", "r")
print(file.read(20))
file.close()
```

Writing to a File

Write New Content (Overwrites file)

```
file = open("notes.txt", "w")
file.write("Python File Handling\n")
file.write("This is Day 20 lesson.")
file.close()
```

Append Content (Add to existing file)

```
file = open("notes.txt", "a")
file.write("\nAdding new line using append mode.")
file.close()
```

Clean a Text File (Remove Spaces)

```
cleaned = []

with open("sample.txt", "r") as f:
    for line in f:
        cleaned.append(line.strip().title())

with open("cleaned_output.txt", "w") as f:
    for city in cleaned:
        f.write(city + "\n")
```